Example 2.7

A 3-fold invitation card is given with measures as in the Fig. 2.25. Find its area.

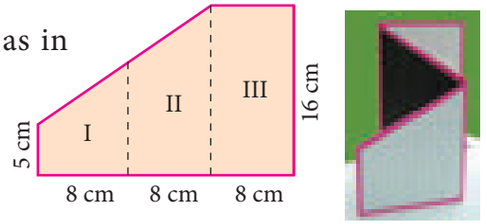


Fig. 2.25

Solution:

Figures I and II are trapeziums separately as well as combinedly.

The parallel sides of the combined trapezium (I and II) are 5 *cm* and 16 *cm* and its height, $h = 8 + 8 = 16\,cm$, length of the rectangle (III) = 16 *cm* and its breadth = 8 *cm*

$\therefore$ Area of the combined invitation card

= area of the combined trapezium + area of the rectangle

$$= \frac{1}{2} \times h \times (a + b) + l \times b$$

$$= \frac{1}{2} \times 16 \times (5 + 16) + 16 \times 8$$

$$= 168 + 128 = 296 \ cm^2$$

Aliter:

Area of the invitation card

= area of the outer rectangle –
area of the right angled triangle

$$= l \times b - \frac{1}{2} \times b \times h$$

$$= 24 \times 16 - \frac{1}{2} \times 11 \times 16$$

$$= 384 - 88 = 296 \ cm^2$$

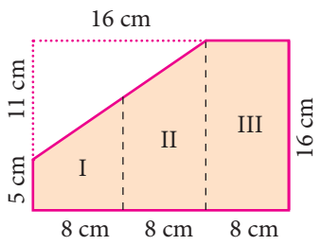


Fig. 2.26

Example 2.8

Seenu wants to buy a floor mat for his kitchen at home as given in Fig. 2.27. If the cost of the mat is ₹ 20 per square foot, what will be the cost of the entire mat?

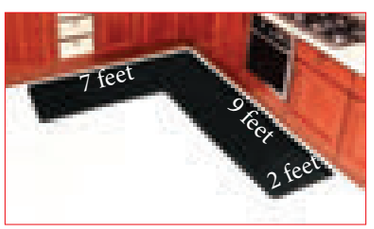


Fig. 2.27

Solution:

The mat given in the figure can be split into two rectangles as follows:

02_8th_Maths_Measurements_Chap 2.indd 63 2/12/2020 12:07:14 PM